DBMS SUMMER WORKSHOP

BRAZIL WON JULES RIMET!
ENGLAND THEFT JULES RIMET!!

A Historical SQL Investigation

Database: MySQL 8.x
Format: Hands-on investigation workshop

THE CASE

London, March 1966.

The FIFA World Cup is only months away.

The Jules Rimet Trophy — the trophy awarded to the winner of the World Cup — is being displayed at the Stampex exhibition at Westminster Central Hall.

On Sunday, **20 March 1966**, something went terribly wrong.

During a routine check around noon, security discovers that:

- The display case has been opened;
- The rear doors have been forced;
- The trophy is gone.

The theft became international news. A ransom demand follows. A suspect is arrested. But, but, but! The trophy is still missing.

Then, seven days after the theft, a dog named **Pickles** discovers a mysterious package under a hedge in South London. Guess what was inside the package!

Let us re-investigate the real historical case that contains enough clues to make an excellent database investigation. Our workshop will focus on reconstructing those clues as a relational database and use SQL to connect them.

Nota Bene

This is not claimed to be the original Metropolitan Police database.

Therefore:

> **Use the database evidence to practice SQL, but do not treat every database row as a historical archival record.**

YOUR ROLE

You are a junior database analyst assisting the investigation. Your job is NOT simply to write SQL. Your job is to answer questions such as:

- Who had access?
- Where did the investigation take place?
- Which evidence belongs to which person?
- What happened before and after the ransom demand?
- Which witnesses observed what?
- Can we reconstruct the sequence of events?

And eventually see: *CAN THE DATABASE HELP US CONNECT THE CLUES?*

LAB 01

CASE FILE 01

THE TROPHY DISAPPEARS

PART 1 — UNDERSTAND THE CASE

Before touching SQL, read the timeline.

Historical Timeline

Date	Event
January 1966	Football Association receives the trophy
March 19	Stampex exhibition opens
March 20	Trophy discovered stolen
March 21	Ransom contact begins
March 24–25	Police attempt ransom operation
March 27	Pickles discovers trophy
March 28	Recovery publicly announced
April 18	Trophy returned to FA

The trophy had been insured for £30,000 despite an official value of £3,000. The exhibition was held at Westminster Central Hall, and security arrangements included uniformed and plainclothes officers, although the trophy was not continuously watched by a guard.

Mission-01

THINK LIKE A DATABASE DESIGNER

What if the entire investigation was stored in single investigation book:

- person_name
- person_role
- phone
- location
- building
- room
- event
- event_date
- event_time
- evidence_description
- witness_name
- witness_statement
- ransom_amount
- police_officer

Think about:

- repeated person information
- repeated locations
- multiple events involving one person
- multiple witnesses
- evidence belonging to different events
- changing information
- duplicate data

YOUR FIRST TASK

Identify entities that should become separate tables. Possible candidates:

- person
- location
- event
- witness_statement
- evidence
- ransom_demand
- police_operation
- trophy

You may discover additional entities.

Mission-02

CREATE THE DATABASE

```
CREATE DATABASE jules_rimet_investigation;

USE jules_rimet_investigation;

SHOW DATABASES;
```

Mission-03

THE TROPHY

Create a table to represent the stolen object.

Requirements:

- unique ID
- name
- type
- official value
- insurance value
- status

```
CREATE TABLE trophy (

    trophy_id BIGINT PRIMARY KEY AUTO_INCREMENT,

    trophy_name VARCHAR(100) NOT NULL,

    trophy_type VARCHAR(100),

    official_value DECIMAL(12,2),

    insurance_value DECIMAL(12,2),
```

```
    status VARCHAR(30) NOT NULL
);
```

Insert the Jules Rimet Trophy:

```
INSERT INTO trophy
(trophy_name, trophy_type, official_value, insurance_value, status)
VALUES
(
    'Jules Rimet Trophy',
    'FIFA World Cup Trophy',
    3000.00,
    30000.00,
    'STOLEN'
);
```

Ask yourself!

Why didn't we store £3,000 as text and used DECIMAL(12,2) instead.

Mission-04

PEOPLE OF THE CASE

```
CREATE TABLE person (
```

```
    person_id BIGINT PRIMARY KEY AUTO_INCREMENT,
    full_name VARCHAR(120) NOT NULL,
    role VARCHAR(80) NOT NULL,
    notes TEXT
);
```

LOAD THE KNOWN PEOPLE! All of them!!

```
INSERT INTO person
(full_name, role, notes)
VALUES
('Joe Mears', 'Football Association Chairman',
 'Chairman of the Football Association'),

('Edward Betchley', 'Ransom Intermediary',
 'Known as Jackson during ransom operation'),

('Charles Buggy', 'Detective Inspector',
 'Flying Squad officer involved in ransom operation'),

('David Corbett', 'Civilian',
 'Recovered the trophy after Pickles discovered it'),
```

```
('Pickles', 'Dog',
 'Discovered the trophy under a hedge'),

('Frank Hudson', 'Security Guard',
 'Worked at the exhibition'),

('Margaret Coombes', 'Witness',
 'Churchgoer who reported seeing a suspicious man'),

('Harold Mayes', 'Football Association Official',
 'Identified the recovered trophy');
```

Remember

Not every person in this table is a suspect. A database row does not mean: person = criminal. The database records a person's **relationship to the investigation**.

Ask yourself!

Does the table name reflect real context?

Mission-05

Add Locations

```
CREATE TABLE location (
    location_id BIGINT PRIMARY KEY AUTO_INCREMENT,
    location_name VARCHAR(150) NOT NULL,
    city VARCHAR(100),
    description TEXT
);
INSERT INTO location
(location_name, city, description)
VALUES
(
    'Westminster Central Hall',
    'London',
    'Exhibition venue where the trophy was displayed'
),
(
    'Battersea Park',
    'London',
    'Location selected for ransom exchange'
),
(
    'Kennington Park Road',
    'London',
```

```
    'Location where the ransom suspect attempted to escape'
),
(
    'Beulah Hill',
    'London',
    'Area where Pickles discovered the package'
),
(
    'Cannon Row Police Station',
    'London',
    'Police station where Corbett handed over the recovered trophy'
),
(
    'Gipsy Hill Police Station',
    'London',
    'Police station where the recovered trophy was identified'
);
```

Mission-06

EVENTS

Now we need to represent what happened. Remember every event happens in a location and we already have a location table. We just need to connect the location id here.

```
CREATE TABLE investigation_event (
    event_id BIGINT PRIMARY KEY AUTO_INCREMENT,
    event_name VARCHAR(150) NOT NULL,
    event_date DATE NOT NULL,
    event_time TIME,
    location_id BIGINT,
    description TEXT,

    FOREIGN KEY (location_id)
        REFERENCES location(location_id)
);
```

INSERT HISTORICAL EVENTS

```
INSERT INTO investigation_event
(event_name, event_date, event_time, location_id, description)
VALUES

(
   'Exhibition Opens',
```

```
    '1966-03-19',
    NULL,
    1,
    'Stampex exhibition opens'
),

(
    'Trophy Handed to Police',
    '1966-03-27',
    NULL,
    4,
    'David Corbett hands trophy to police'
),

(
    'Trophy Discovered Missing',
    '1966-03-20',
    '12:10:00',
    1,
    'Display case found opened and trophy missing'
),
(
    'Routine Security Check',
```

```
    '1966-03-20',
    '11:00:00',
    1,
    'Trophy observed before the theft window'
),

(
    'Ransom Meeting',
    '1966-03-25',
    NULL,
    2,
    'Police operation involving alleged intermediary Jackson'
),

(
    'Suspect Escape Attempt',
    '1966-03-25',
    NULL,
    3,
    'Suspect attempts to escape from police operation'
),

(
```

```
    'Trophy Discovered',
    '1966-03-27',
    NULL,
    5,
    'Pickles discovers wrapped package'
),

(
    'Ransom Contact',
    '1966-03-21',
    NULL,
    NULL,
    'Anonymous contact and ransom demand'
);
```

Ask Yourself

- Could event time logging be differently modelled? Like Timestamp? Reasoning.
- What would happen if we mistakenly entered an event log with an unregistered location?
- What would happen if we mistakenly deleted or updated location_id of a location that has events registered?

Mission-07

YOUR FIRST INVESTIGATION QUERY

The investigation team asks:

Show every event that occurred at Westminster Central Hall.

Try:

```
SELECT *

FROM investigation_event

WHERE location_id = 1;
```

But the investigator doesn't want to see ids, they want to see human readable data only. Also, they don’t remember location_id = 1 means Westminster Central Hall, right? So, it is our job to connect the location identities.

JOIN THE EVIDENCE

```
SELECT

    e.event_name,

    e.event_date,

    e.event_time,

    l.location_name,

    e.description

FROM investigation_event e

JOIN location l

    ON e.location_id = l.location_id;
```

Help the investigating officer

Find:

- All events that happened in London.

- All events having the trophy written in the event notes.
- All events after 20 March 1966.
- All events are ordered chronologically.

Mission-08

WITNESS STATEMENTS

The police interviewed witnesses and maintained records. We need to understand that every witness to an event is a person meaning every statement has a relation with person record and event record.

```
CREATE TABLE witness_statement (

    statement_id BIGINT PRIMARY KEY AUTO_INCREMENT,

    person_id BIGINT NOT NULL,

    event_id BIGINT,

    statement_date DATE,

    statement_text TEXT,



    FOREIGN KEY (person_id)

        REFERENCES person(person_id),



    FOREIGN KEY (event_id)

        REFERENCES investigation_event(event_id)

);
```

```
INSERT INTO witness_statement

(person_id, event_id, statement_date, statement_text)

VALUES



(

    6,

    3,

    '1966-03-20',

    'Reported that a suspicious man had been seen near the public
telephone area.'

),



(

    7,

    3,

    '1966-03-20',

    'Reported seeing a strange man in the Central Hall area.'

);
```

The historical accounts differ in the descriptions provided by witnesses, which is itself useful for discussing how databases represent conflicting evidence.

Mission-09

WHO SAID WHAT?

Write a query showing the witness intro and his statement on an event. You will need to connect a person to witness statements to event records.

Mission-10

More Witness List

The investigation team asks:

> **Which people in our database have never given a witness statement?**

```
SELECT
    p.full_name,
    ws.statement_id
FROM person p
LEFT JOIN witness_statement ws
    ON p.person_id = ws.person_id;
```

Can you identify the people with no statement? They are your next callees.

Ask Yourself

Why did we use LEFT JOIN instead of INNER JOIN?

Mission-11

COUNT THE EVIDENCE

Which person is associated with the greatest number of investigation records?

```
SELECT
    person_id,
    COUNT(*) AS total_records
FROM witness_statement
GROUP BY person_id;
```

Now make it human-readable by joining a person.

```
SELECT
    p.full_name,
    COUNT(*) AS total_records
FROM witness_statement s JOIN person p
ON s.person_id = p.person_id
GROUP BY p.full_name;
```

YOUR TURN

- Find a summary of events per location. List location names and events occurred at that location by joining and grouping location and event records.
- Find the location where max events were recorded.
- Find the person associated with the highest number of witness statements.
- Find the person associated with the second highest number of witness statements.

LAB 01 MINI CASE

The investigation team wants to reconstruct the beginning of the incident.

Produce a report showing:

- Event
- Date
- Time
- Location
- Description

ordered chronologically

```
SELECT
    e.event_name,
    e.event_date,
    e.event_time,
    l.location_name,
    e.description
  from investigation_event e JOIN location l
    ON e.location_id = l.location_id
  ORDER BY  e.event_time DESC;
```

Then answer: **What is the last documented event before the trophy was discovered missing?**

```
SELECT
    e.event_name,
    e.event_date,
    e.event_time,
    l.location_name,
    e.description
  from investigation_event e JOIN location l
    ON e.location_id = l.location_id
  WHERE e.event_date < (
      SELECT event_date
        from investigation_event
        WHERE e.event_name = "Trophy Discovered Missing"
  )
  ORDER BY  e.event_time DESC;
```

Do It Yourself

Find out: **What is the first documented event after the trophy was discovered missing?**

Mission-12

The Case File

```
CREATE VIEW v_case_file AS
SELECT
    s.statement_date,
    s.statement_text,
    e.event_name,
    e.event_date,
    e.event_time,
    l.location_name,
    e.description
FROM witness_statement s
JOIN investigation_event e
    ON s.event_id = e.event_id
JOIN location l
    ON e.location_id = l.location_id;
```

Now:

```
SELECT *
FROM investigation_timeline
ORDER BY event_date, event_time;
```

Ask Yourself

Why would an investigator prefer the view over rewriting the JOIN every time?

Do It Yourself

- Can you create a witness leader board for the investigators?

Mission-13

Case Map

The investigation director wants a precomputed timeline summary.

```
CREATE TABLE mv_case_map AS

SELECT

    e.event_id,

    e.event_name,

    e.event_date,

    e.event_time,

    l.location_name,

    e.description

FROM investigation_event e

LEFT JOIN location l

    ON e.location_id = l.location_id;
```

Now:

```
SELECT *

FROM investigation_timeline

ORDER BY event_date, event_time;
```

Do It Yourself

- Frank Hudson changes his statement that it was not the telephone area, rather it was the reception area.
- At some point the location of the ransom contract was found to be **Battersea Park**. Update the investigation_event table with the id of this location and check the case map again.
- Check your case file and case map again and share your thoughts.

This is where the case had a new turn! See you in the next episode!!